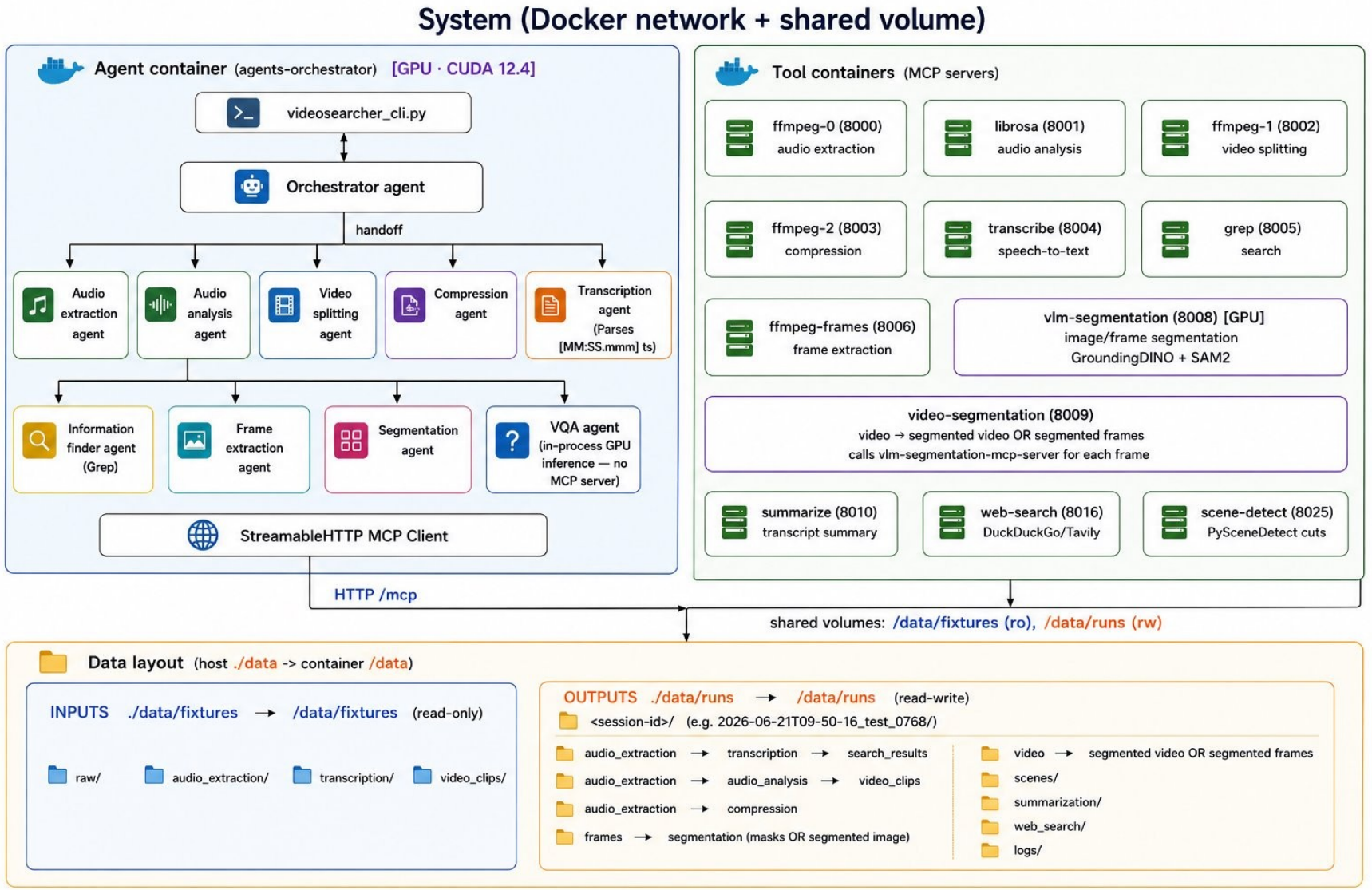
VideoSearcher AgenticAI

Orchestrator LLM benchmark & sampling optimization

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System overview



Benchmark setup

Model	Size	Architecture
qwen3.6-35B-A3B	35B	MoE
gemma-4-12b	12B	dense
gemma-4-E2B	5B	dense
qwen3.6-27B	27B	dense
gpt-oss-120b	120B	MoE

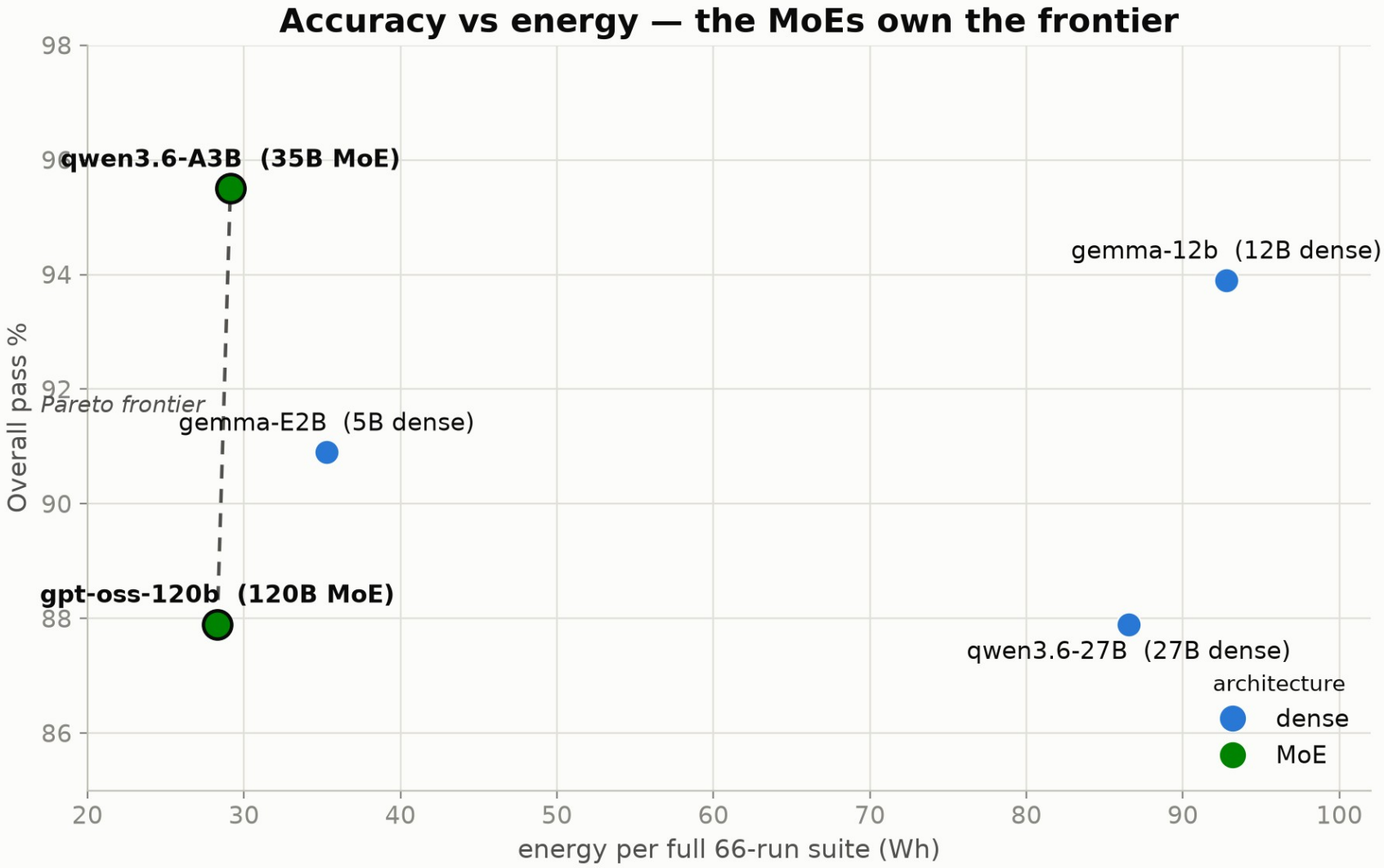
- **14** metrics recorded per run
- Defaults sampling; full **66**-run suite per model
- Hardware: DGX GB10, **128 GB** unified memory

Benchmark results — 66-run suite, defaults sampling

Model	Overall %	Handoff %	Vision %	Result %	Energy (Wh/suite)	passes/kWh
qwen3.6-35B-A3B (35B MoE)	**95.5**	95.5	**100**	**100**	29.1	**2163**
gemma-4-12b (12B dense)	93.9	**97.0**	**100**	92.6	92.8	668
gemma-4-E2B (5B dense)	90.9	93.9	88.9	92.6	35.3	1701
qwen3.6-27B (27B dense)	87.9	89.4	**100**	96.3	86.5	670
gpt-oss-120b (120B MoE)	87.9	87.9	88.9	96.3	28.3	2050

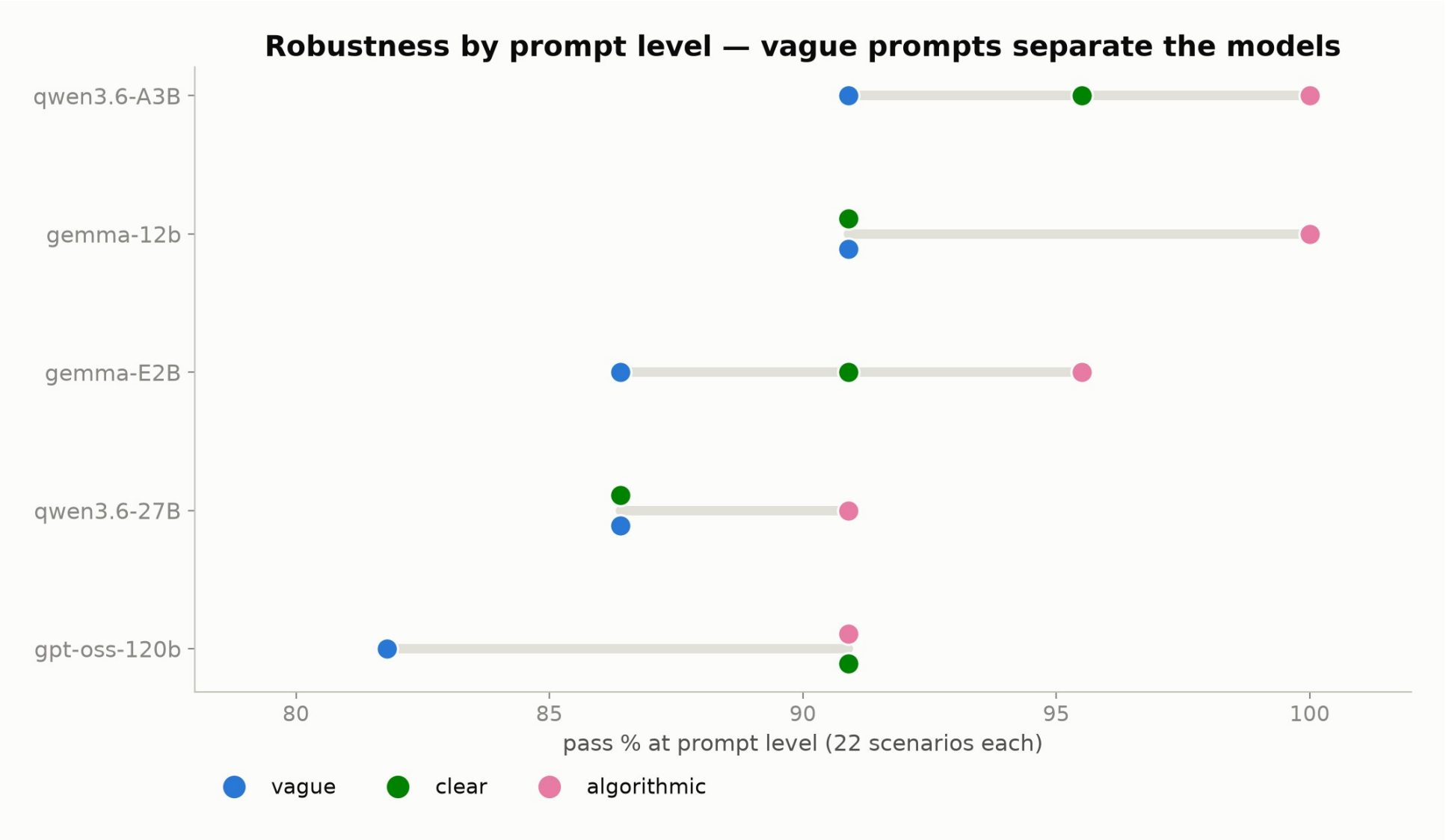
- MoEs lead accuracy and energy efficiency
- Dense models pay ~**3×** energy per suite

Accuracy vs energy — Pareto frontier



- MoEs own the accuracy–energy Pareto frontier
- a3b: **95.5%** at **29.1 Wh** → **2163** passes/kWh
- Dense models pay **~3×** energy

Robustness by prompt level

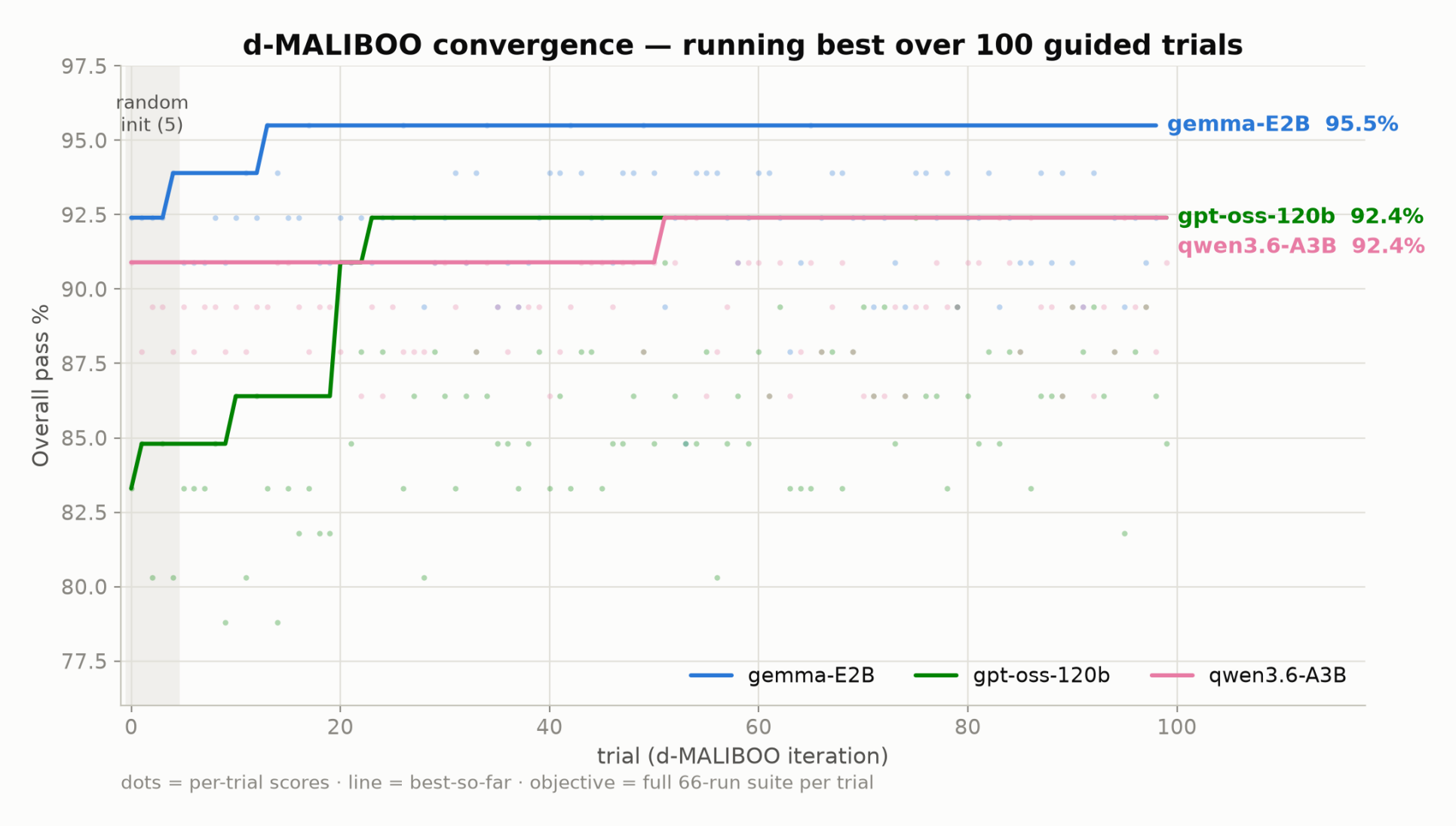


- Algorithmic prompts near-saturated
- Vague prompts separate the models
- gpt-oss: **81.8%** on vague

Sampling tuning — d-MALIBOO methodology

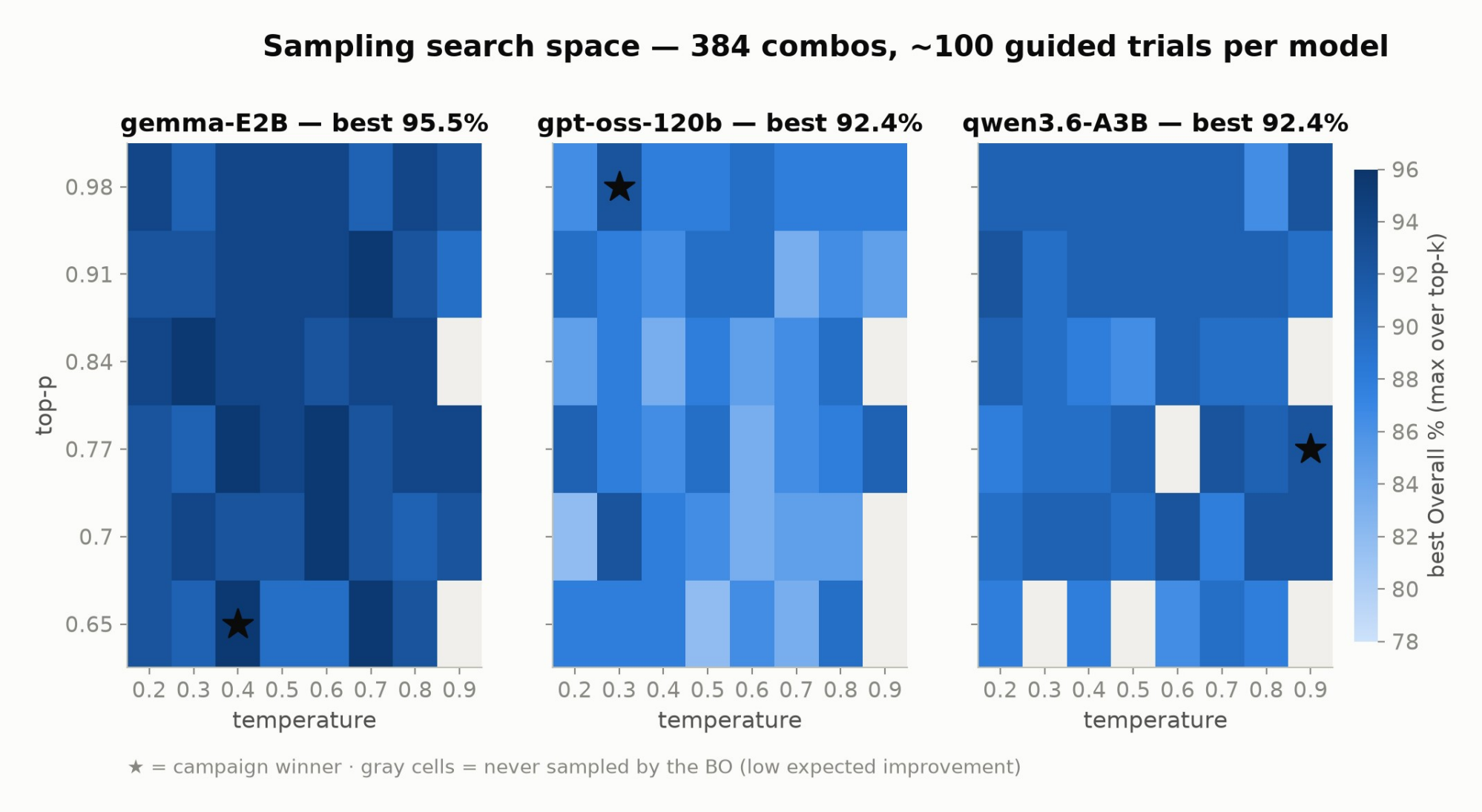
- Grid: temperature × top-p × top-k = **384** combos
- t {0.2..0.9} · p {0.65, 0.70, 0.77, 0.84, 0.91, 0.98} · k {10..80}
- Budget: **5** random init + **95** guided trials per model
- d-MALIBOO (PoliMI): Gaussian-Process surrogate, Matérn kernel
- Expected Improvement augmented by ML model (MLP classifier)
- Trials guided, not random; objective = suite Overall %

Convergence of the guided search



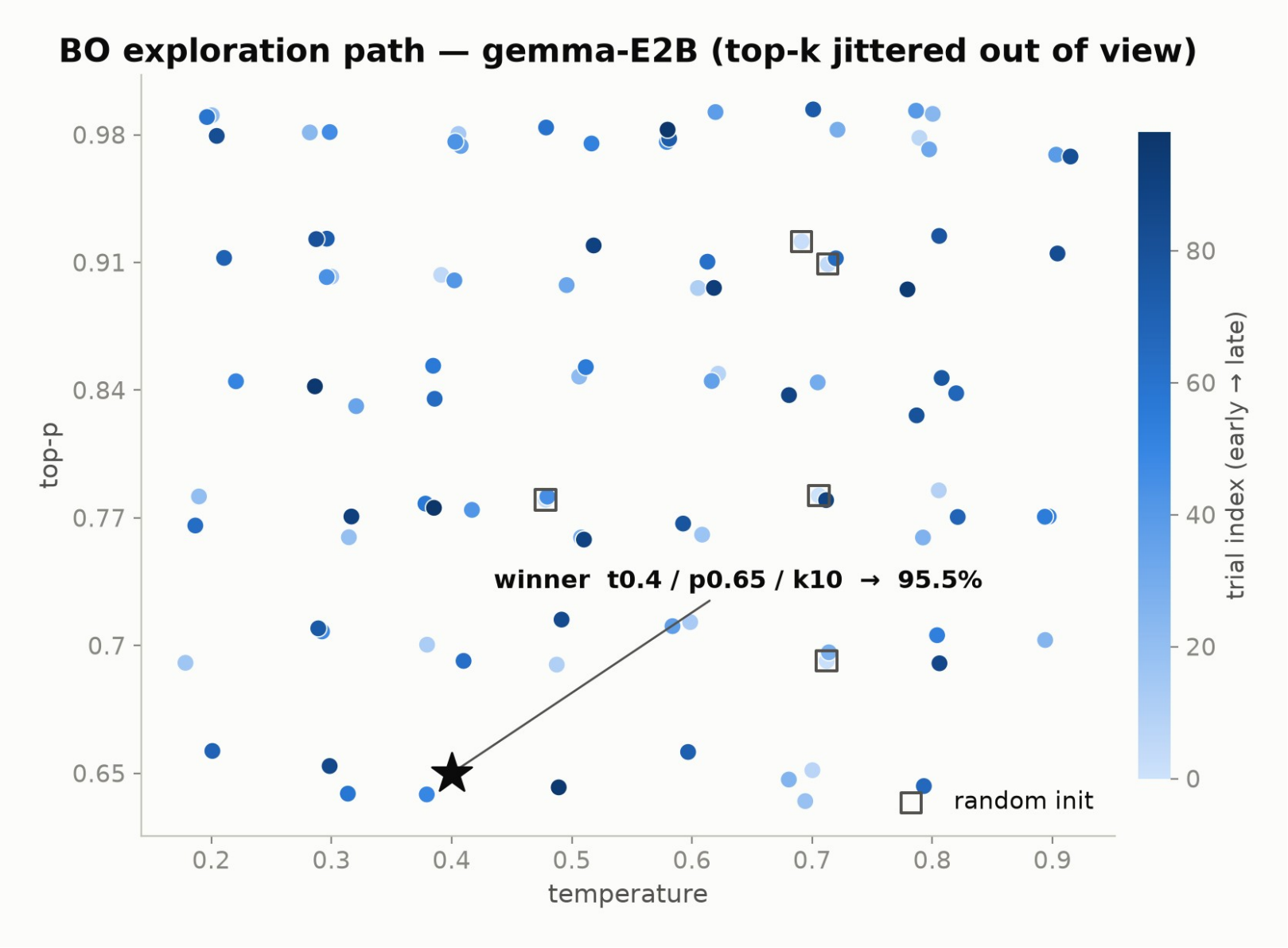
- **5** random init, then guided trials
- Plateaus reached within ~**20–50** trials

Search-space landscape



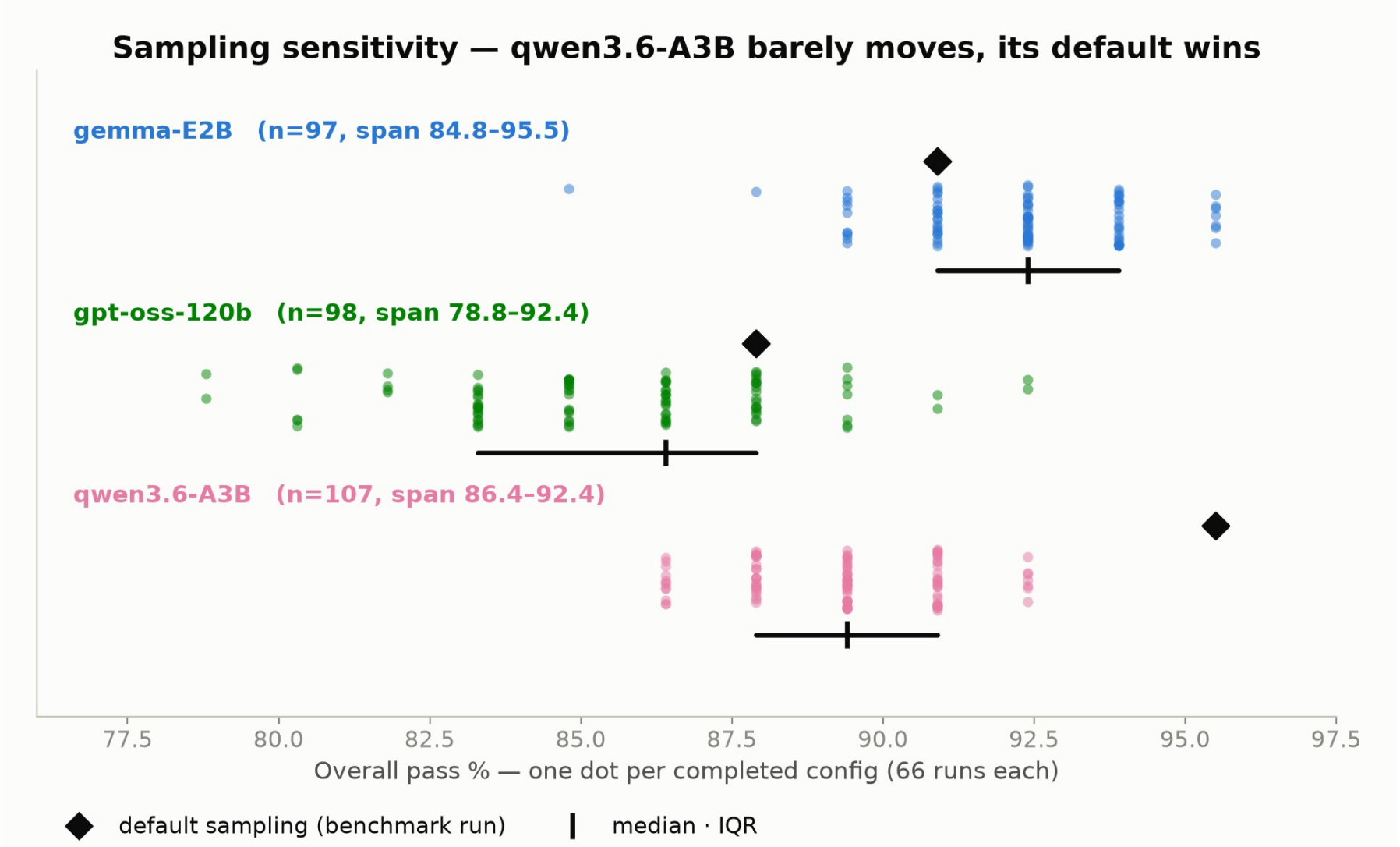
- gemma: broad low-temp / low-k plateau
- a3b: flat
- gray cells: never sampled

Exploration path — winner model



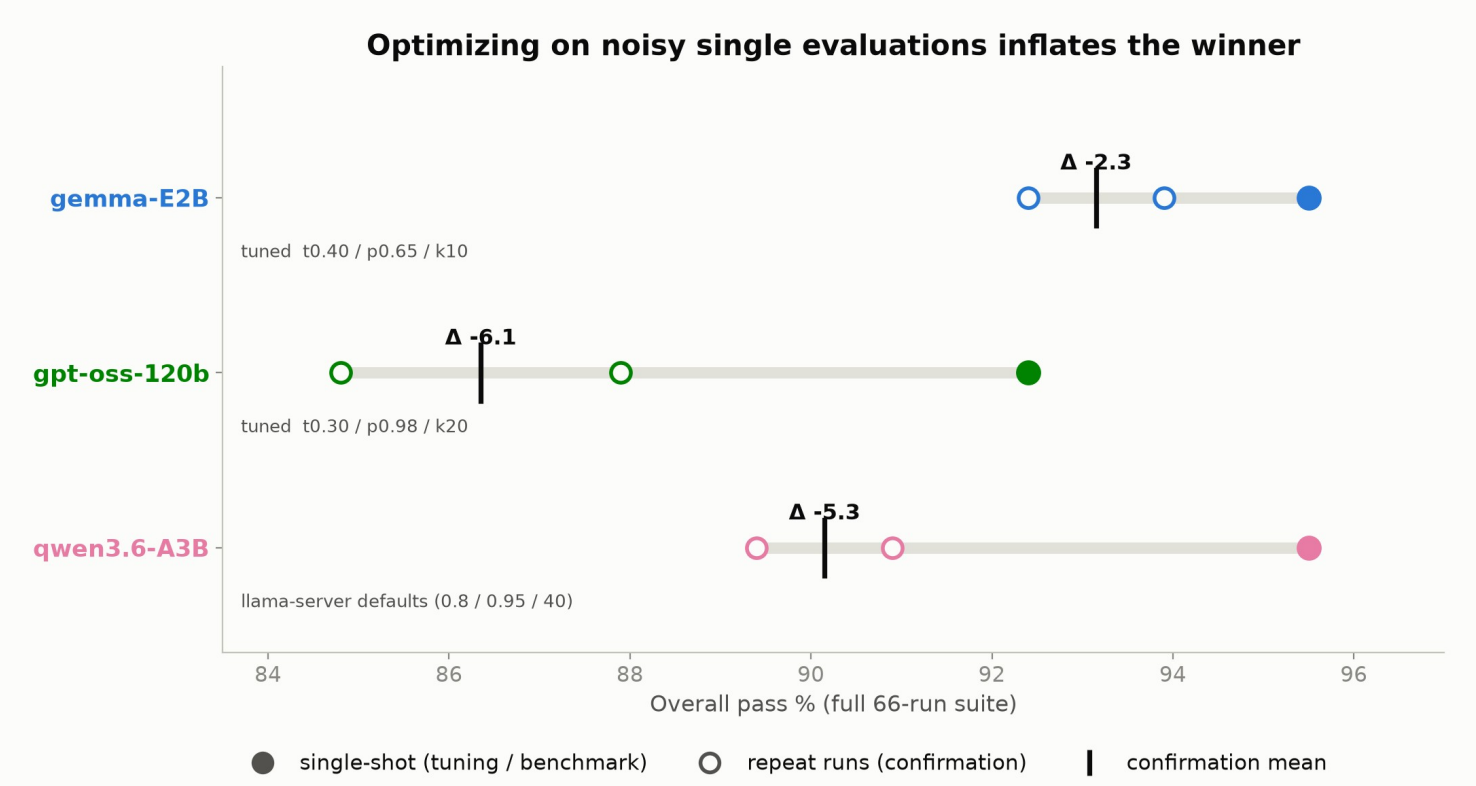
- Exploration → exploitation on the winner model
- Random-init squares; early→late color ramp
- Winner starred

Sampling sensitivity



- gpt-oss spans **13.6** pts — sampling matters
- a3b spans **6** (**107** configs, **86.4–92.4**)
- a3b default **95.5** wins → tuning not useful

Noise & optimizer bias — confirmation runs



- Single-shot winners inflated **2.3–6.1** pts
- Run-to-run spread **1.5–3.1** pts
- Report confirmation means

Config	single-shot	repeats	mean	Δ
gemma-E2B t0.40/p0.65/k10	95.5	93.9 · 92.4	**93.2**	**−2.3**
gpt-oss t0.30/p0.98/k20	92.4	84.8 · 87.9	**86.3**	**−6.1**
qwen3.6-A3B defaults	95.5	89.4 · 90.9	**90.2**	**−5.3**

Conclusions & future work

- gemma-E2B: **t0.40 / p0.65 / k10** — confirmed mean **93.2%**
- gpt-oss-120b: **t0.30 / p0.98 / k20** — confirmed mean **86.3%**
- qwen3.6-A3B: keep defaults — best observed **95.5%**
- MoE models for energy-bounded deployments
- Limitation: noisy objective → single-shot winners optimistically biased
- Future work: **k≥3** averaging per trial, noise-aware GP, larger fixture set